

FICHTELBERG, Germany

WMO: 105780

Lat: 50.4283N

Lon: 12.9536E

Elev: 1215

StdP: 87.56

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.1	-14.5	-24.6	0.5	-6.7	-20.5	0.7	-7.7	25.1	0.1	23.1	-0.7	7.8	90	1.049

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	21.5	14.9	19.8	14.0	18.2	13.2	15.7	19.9	14.7	18.3	13.8	17.1	5.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	11.7	17.1	13.2	11.0	15.9	12.5	10.4	15.2	48.4	20.1	45.3	18.4	42.8	17.2	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	19.7	17.8	DB	-17.9	25.4	5.0	2.0	-21.5	26.9	-24.5	28.0	-27.3	29.1	-31.0	30.5	(4)
(5)			WB	-18.1	17.9	5.0	0.9	-21.7	18.5	-24.6	19.0	-27.4	19.5	-31.0	20.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-4.3	-5.0	-1.9	1.4	6.8	9.6	12.1	12.3	8.5	4.5	-0.4	-2.5	(6)	
	DBStd	7.44	4.97	5.42	4.38	4.27	4.16	4.05	3.88	3.87	3.72	4.28	4.42	4.06	(7)	
	HDD10.0	2655	442	419	368	258	114	57	23	22	73	177	313	389	(8)	
	HDD18.3	5426	701	653	626	507	358	263	194	191	294	429	563	647	(9)	
	CDD10.0	276	0	0	0	1	15	44	89	92	28	6	0	0	(10)	
	CDD18.3	6	0	0	0	0	0	1	2	3	0	0	0	0	(11)	
	CDH23.3	17	0	0	0	0	0	1	9	7	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	0	1	1	0	0	0	0	(13)	
(14) Wind	WSAvg	9.1	11.7	10.1	9.5	8.0	7.5	7.4	7.1	7.2	8.7	10.2	10.5	11.1	(14)	
(15) Precipitation	PrecAvg	1149	94	79	88	75	97	110	120	109	95	79	94	109	(15)	
	PrecMax	1683	243	206	242	185	198	242	245	299	236	169	219	276	(16)	
	PrecMin	769	13	13	24	8	31	51	24	20	16	5	1	7	(17)	
	PrecStd	215	49	47	47	36	44	38	49	54	47	41	47	56	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	8.5	11.2	13.8	18.1	21.6	24.6	24.2	19.5	16.5	11.9	8.3	(19)	
		MCWB	1.8	2.7	5.4	7.4	11.1	14.8	16.5	16.3	13.5	10.4	5.1	2.5	(20)	
	2%	DB	4.3	6.2	7.8	11.8	16.1	19.2	21.6	21.7	17.5	14.0	9.6	6.1	(21)	
		MCWB	0.0	2.3	4.5	6.9	10.5	13.7	14.8	15.2	12.8	8.6	4.4	2.2	(22)	
	5%	DB	2.8	4.3	5.6	9.8	14.3	17.3	19.9	19.7	15.7	12.1	7.6	4.5	(23)	
		MCWB	0.2	1.5	2.9	5.7	9.7	12.7	14.0	14.2	11.8	8.2	4.3	1.4	(24)	
	10%	DB	1.5	2.4	3.7	7.8	12.8	15.6	18.0	17.9	13.9	10.1	5.7	3.2	(25)	
		MCWB	-0.3	0.0	2.2	4.9	8.8	11.6	13.2	13.3	11.0	7.2	3.4	1.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	4.8	7.4	8.9	12.7	15.8	17.2	17.2	14.6	11.9	7.1	5.6	(27)	
		MCDB	4.9	6.7	9.9	12.3	16.2	19.7	22.4	22.7	17.6	14.9	8.3	6.0	(28)	
	2%	WB	2.2	3.3	4.9	7.5	11.3	14.5	15.8	15.8	13.5	10.4	6.0	3.9	(29)	
		MCDB	2.8	4.6	6.7	10.4	14.4	17.7	20.0	20.3	16.2	12.5	7.4	4.7	(30)	
	5%	WB	1.2	1.7	3.4	6.4	10.3	13.4	14.7	14.7	12.5	9.1	5.1	2.6	(31)	
		MCDB	1.7	3.6	5.0	9.0	13.5	16.3	18.3	18.6	14.9	10.5	7.0	3.4	(32)	
	10%	WB	0.1	0.3	2.3	5.1	9.3	12.3	13.7	13.8	11.5	8.2	3.9	1.6	(33)	
		MCDB	0.9	1.7	3.6	7.4	12.3	15.0	17.1	17.0	13.5	9.5	5.4	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.5	4.5	5.5	6.4	6.1	6.7	6.3	5.5	4.9	4.2	4.1	(35)	
		MCDBR	6.0	6.0	6.8	7.9	8.4	8.0	9.3	8.5	7.6	7.3	6.2	5.8	(36)	
	5% WB	MCWBR	4.6	4.3	4.7	4.8	4.8	4.8	5.1	4.9	4.7	4.6	4.5	4.0	(37)	
		MCWBR	5.2	5.3	6.3	7.4	7.7	7.4	8.1	7.6	6.8	6.3	5.5	4.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.259	0.277	0.311	0.343	0.346	0.346	0.357	0.351	0.325	0.305	0.282	0.256	(40)	
		taud	2.455	2.440	2.426	2.386	2.394	2.403	2.368	2.404	2.467	2.523	2.532	2.476	(41)	
	Edn at Noon	Edn at Noon	806	876	899	901	910	910	893	882	872	831	771	757	(42)	
		Edh at Noon	57	76	93	108	112	112	115	105	88	70	54	49	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.81	1.61	2.58	4.03	4.83	5.24	4.99	4.34	3.10	1.85	0.88	0.63	(44)	
	RadStd	RadStd	0.15	0.24	0.37	0.53	0.59	0.45	0.54	0.42	0.43	0.24	0.11	0.08	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (51 neighbors)	+0.40	N/A	N/A	+0.32	+0.35	N/A	N/A	+52	+16	(47)

Nomenclature: See separate page